

HAT-MedMNIST

A hybrid agentic training framework for MedMNIST

Stefano Verrilli

Project Manager

scope, schedule, resources, costs,
risk management and reporting

Giuseppe Todisco

Technical Manager

architecture, agent design,
representation and training strategy,
integration

Mohammad Dolatabadi

Reviewer

independent review of KPIs,
traceability, reproducibility and
TRL; the WP6 reviewer agent

Agentic AI / medical-image benchmarks

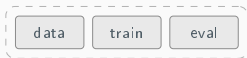
- 1 **Motivation, Objectives & Plan**

- 2 **System Architecture & MVP**

- 3 **Prior Art, Governance & Validation**

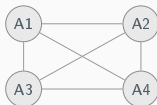
Why HAT-MedMNIST

Script pipeline



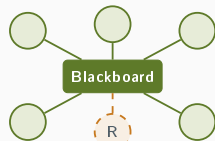
tightly coupled
hard to audit

Typical LLM agents



random at every step
locked to one vendor

HAT-MedMNIST



typed contracts · provenance
independent reviewer with veto

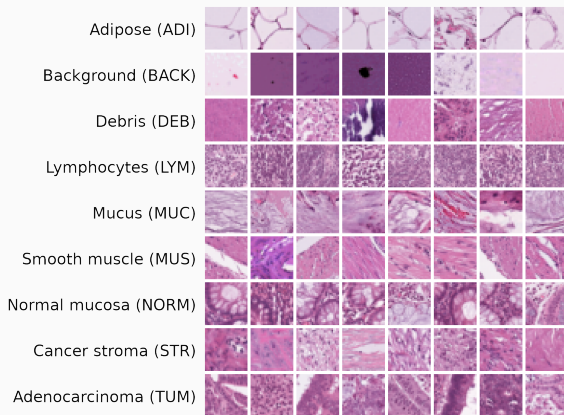
The problem

- Training pipelines are monolithic scripts: hard to audit and to reproduce.
- LLM-agent frameworks let a model improvise every step, tied to one vendor.

Our answer

- One duty per agent; agents exchange only versioned artefacts.
- LLM reasoning only at declared decision points, with a fallback.
- An independent reviewer audits every stage and can veto.

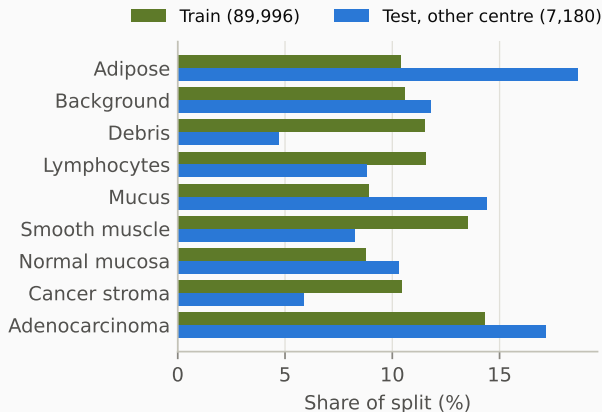
The task in pictures: PathMNIST



Can software agents learn to tell colon tissue types apart?

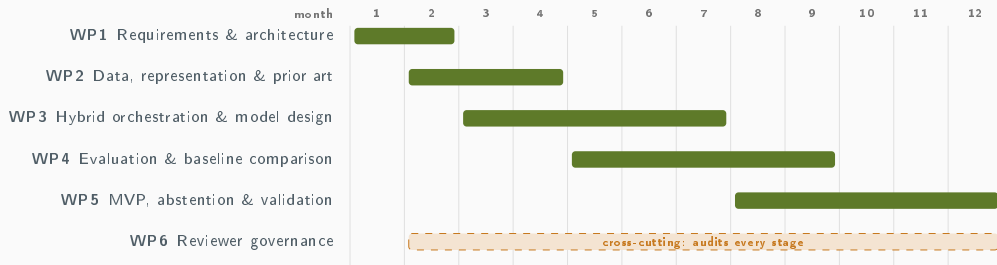
- PathMNIST: stained (H&E) patches from colorectal cancer slides, part of MedMNIST v2.
- 9 tissue classes, 107,180 images of only **28×28 pixels**.
- Train/validation from one source; test from **a different hospital**.
- Small images keep experiments fast; the second hospital keeps the test honest.

The data at a glance



- Official split: 89,996 / 10,004 / 7,180 images.
- Training classes are near-balanced (largest/smallest = 1.63).
- The test hospital differs: adipose 18.6% vs 10.4%, debris 4.7% vs 11.5%.
- So test accuracy measures **transfer** to new data, not memory.

The plan: six work packages in twelve months



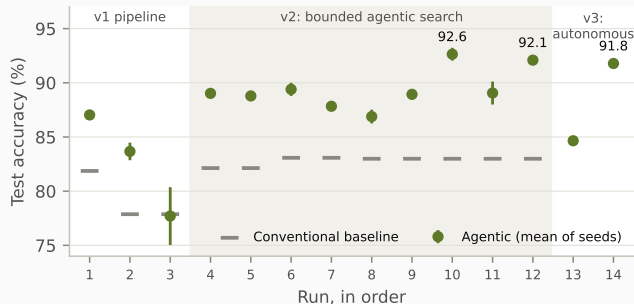
- Hybrid Agile / stage-gate: short iterations, a formal gate at the end of each WP.
- Each WP delivers a reviewable output (D1–D6) with numeric KPIs.
- WP6, the reviewer, runs alongside every other WP.
- Goal: an integrated MVP demonstrated at **TRL 7** in month 12.

Risk map

Impact	High		Leakage Contracts Below baseline	
	Medium	Vendor lock-in	LLM randomness	
	Low			
		Low	Medium	High
		Probability		

- **Leakage**: an audit gate blocks training until splits and statistics are checked.
- **Contracts**: typed artefacts fail fast; the orchestrator can veto.
- **Below baseline**: baseline-first, ablations, negative results reported.
- **LLM randomness**: temperature 0, logged provenance, fallback.
- **Lock-in**: any open model behind one interface.

From first run to MVP: 14 runs, three versions



- **v1**: first end-to-end runs; one tie with the baseline.
- **v2**: agents choose within a bounded search; every run beats the baseline, best 92.6%.
- **v3**: fully autonomous; agents write and launch their own experiment code.
- Same official split throughout, so the runs are comparable.

The flagship run in numbers

3 h 31 min

end-to-end, full dataset

8

model trials, chosen by
agents

28

LLM decisions, local model

72

versioned artefacts

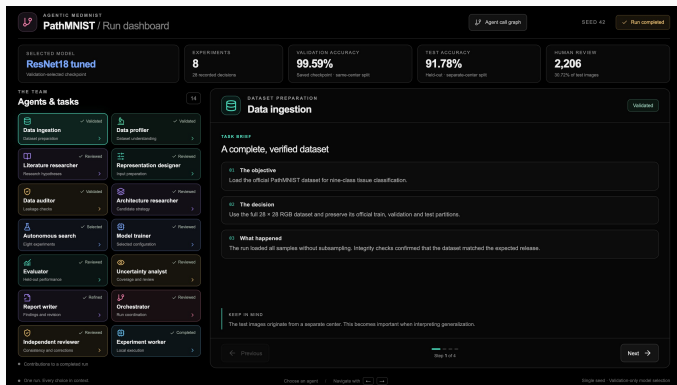
What ran

- 11 stages on the full dataset, one local CUDA machine.
- LLM: qwen3.8 served locally by Ollama; no cloud model.
- The search agent proposed and launched all 8 trials.

Where the time went

- Model search: 3 h 20 min, 95% of the run.
- All other stages together: about 11 minutes.
- 56 recorded calls: 12 stage starts, 12 reviews, 32 worker jobs.

Run explorer: the run at a glance



- One page per run: chosen model, experiments, validation and test accuracy, human-review load.
- 14 agent cards; each opens the objective, the decision and what happened.
- Built for non-specialists: no code, prompts or raw files on screen.
- Opens locally from the stored results; no training or LLM needed.



Done

- Autonomous end-to-end run on the full dataset (v3).
- 3 seeds, conventional baseline, ablations and live arXiv sources (v2).

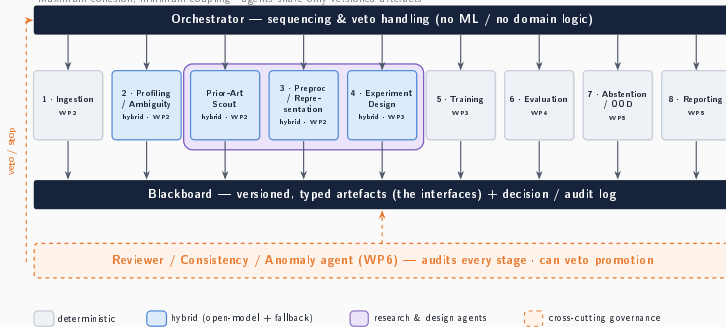
Evidence still to file

- Fault-injection report on the current code.
- Replay of the flagship run; signed mitigation list for reviewer notes.
- Demonstration by someone other than the implementer.

The Blackboard architecture

Hybrid Agentic MedMNIST — architecture

Maximum cohesion, minimum coupling · agents share only versioned artefacts



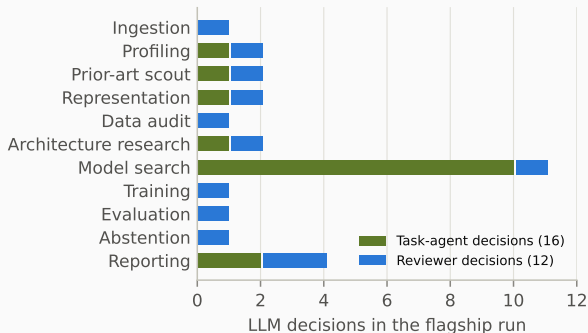
Coordination

- **Orchestrator:** sequences stages; holds no domain logic.
- **Blackboard:** versioned, typed, checksummed artefacts plus a decision log.

Control

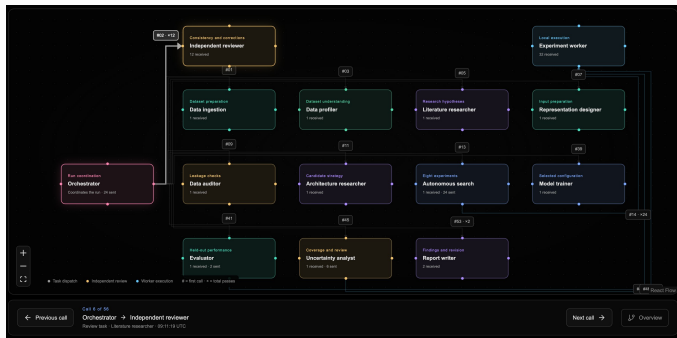
- **Reviewer:** audits after every stage; can request a retry or veto.
- **0 direct agent calls**, verified on the code's syntax tree.

Where the LLM is allowed to decide



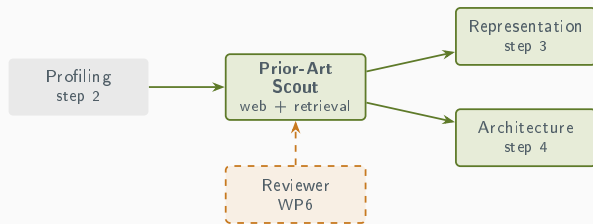
- **Deterministic:** ingestion, data audit, training, evaluation, abstention. No task-agent LLM call.
- **Judgement:** ambiguity, prior art, representation, architecture, search, report.
- Model search uses most decisions: 10 of 16.
- Every decision is logged with its prompt and source; the heuristic fallback was used once.

Run explorer: agent call graph



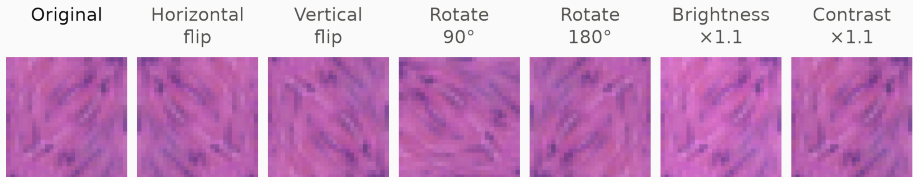
- One box per agent; 56 recorded calls grouped into 15 connections.
- Calls go through the orchestrator, the reviewer or the worker: **no agent calls another**.
- Step through calls in time order, with timestamps and task context.
- 12 stage dispatches, 12 reviews, 32 worker jobs.

The Prior-Art Scout



- Searches arXiv; every source is snapshotted and checksummed.
- Writes a cited brief that feeds representation and architecture.
- Ranks existing options, or proposes new ones that wait at a review gate.
- Advises only: never trains, selects or calls another agent.
- Live sources in the v2 three-seed run: 6 of 6.

Representation: what the model sees



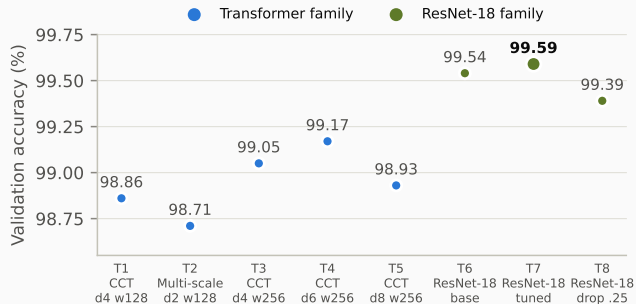
Normalisation

- Per-channel standardisation.
- Statistics from **training images only**, reused unchanged for validation and test.

Augmentation

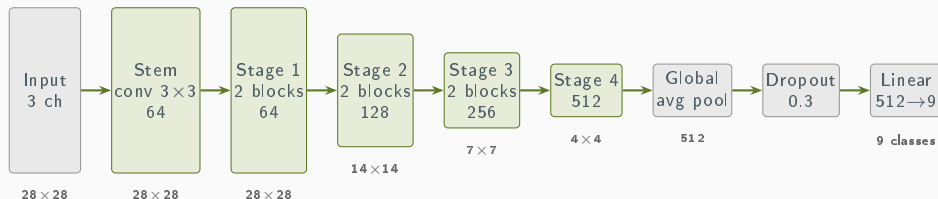
- 6 transforms picked by the representation agent from a label-preserving whitelist.
- Tissue has no fixed orientation, so flips and rotations keep the label.
- 89,996 images → 629,972 training views.

Autonomous model search



- The agent reads every past result before choosing the next trial.
- Trials 1–5: compact and multi-scale transformers, best 99.17%.
- Trial 6: switch to ResNet-18, immediately better (99.54%).
- Selected on validation accuracy, macro-F1 as tie-break. **Test never seen.**

The winning architecture: ResNet-18 for 28×28



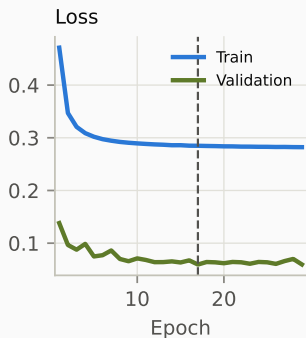
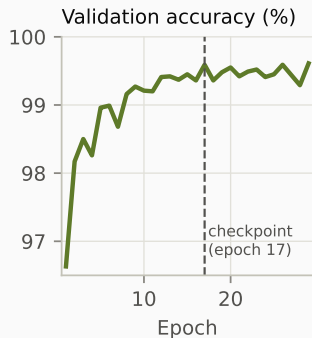
Network

- CIFAR-style ResNet-18: 3×3 stem and **no max-pool**, so small images keep detail.
- 4 stages $\times$ 2 residual blocks, widths $64 \rightarrow 512$.
- **11.17 M** parameters, trained from scratch.

Training recipe

- AdamW, lr 2×10^{-4} , weight decay 0.01, cosine schedule.
- Class-weighted loss, label smoothing 0.05, dropout 0.3.
- Batch 256; early stopping with patience 12.

Training dynamics



- Validation accuracy passes 99% by epoch 5.
- Best checkpoint: epoch 17, **99.59%**.
- Early stopping at epoch 29, after 12 epochs without improvement.
- Train loss stays near 0.28 because of label smoothing and augmentation, not underfitting.

MVP result on the held-out hospital

True class	ADI	BACK	DEB	LYM	MUC	MUS	NORM	STR	TUM
	92					7			
		100							
		3	93			5			
				100					
	4	2			92		1		
		10	5			82		2	1
				1			97		2
		1	13			17	2	52	14
Predicted class	ADI	BACK	DEB	LYM	MUC	MUS	NORM	STR	TUM
				2			2		96

91.8%

test accuracy

0.981

macro AUC

0.890

macro-F1

- 7,180 images from a hospital never seen in training; 95% CI 91.1–92.4%.
- 7.8 points below validation: the cost of changing hospital.
- Lymphocytes and background are perfect; cancer stroma is the weakest row.

Leakage-safe by design



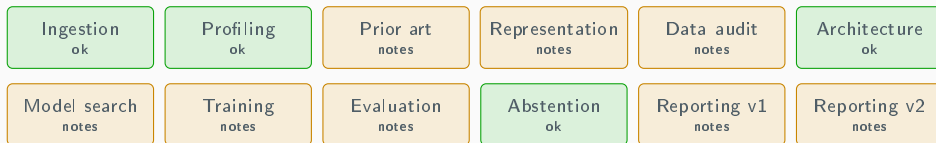
Rules

- Normalisation fitted on training images only.
- Model selection on validation only.
- Test used once, after the model is frozen.

Evidence in the run

- **11/11** audit checks passed before training.
- **0** images shared between train and validation.
- **0** test metrics used in selection.

The reviewer at work



ok = no issues notes = minor issues logged for the record

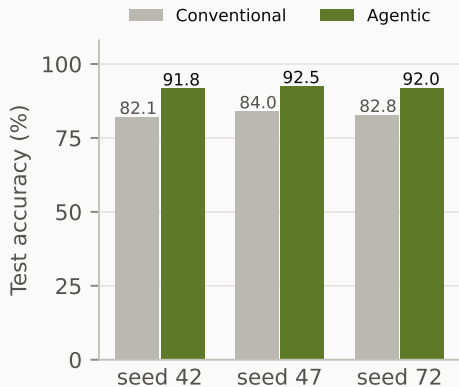
What it checks

- Consistency and arithmetic of every artefact.
- Leakage, provenance and citation validity.
- Written claims against the actual numbers.

What it did

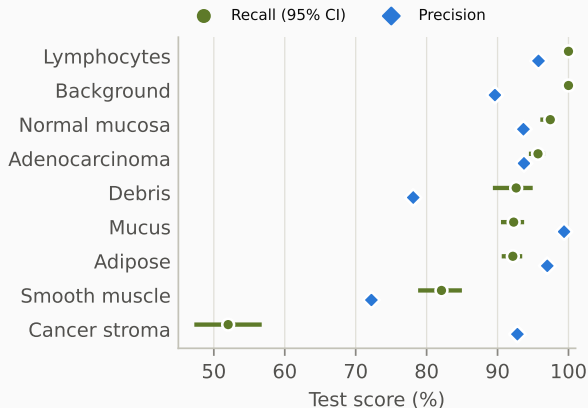
- 12 reviews: 4 clean, 8 with minor notes, **0 vetoes**.
- Caught a rationale claiming 16 image tokens instead of 49.
- Requested a rewrite of the final report (v1 → v2).

Agents vs a conventional pipeline



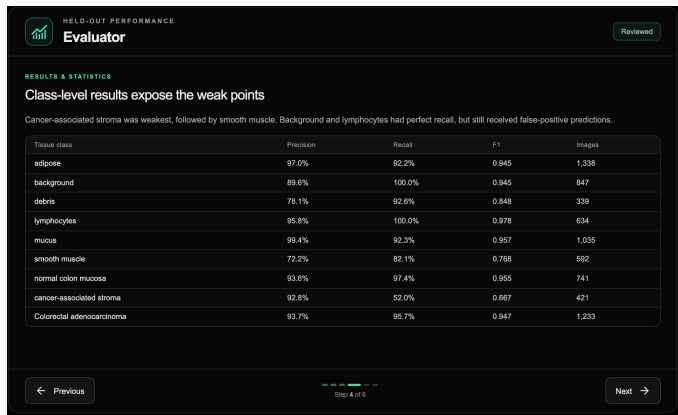
- Same data, splits and seeds; the baseline is a fixed small-CNN recipe.
- Agentic **92.1%** vs conventional 83.0%: **+9.1 points**, on every seed.
- Ablation: representation off 80.9% → standardise 86.3% → + flip 91.3%.
- Evidence: v2 three-seed run, seeds 42/47/72, live arXiv sources.

Error analysis: which tissues fail, and why



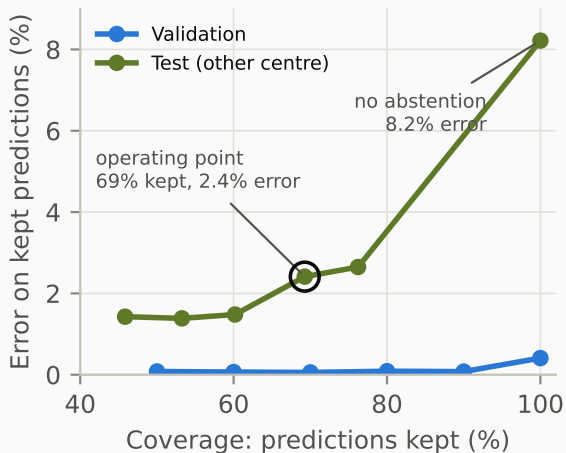
- 7 of 9 tissues reach $\geq 92\%$ recall at the new hospital.
- **Cancer stroma**: 52% recall but 93% precision. Missed cases go to muscle (17%), adenocarcinoma (14%), debris (13%).
- At 28 pixels these tissues share fibrous texture.
- The profiling agent flagged this risk **before training**.

Run explorer: auditing the evaluation



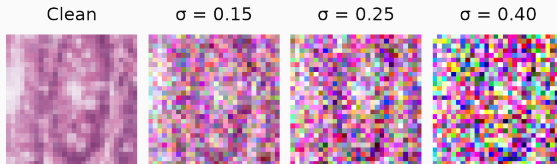
- Per-class precision, recall and F1, exactly as the evaluator agent computed them.
- Every statistic is read from the versioned artefacts, so anyone can re-check it.
- The weak class, cancer stroma, sits next to the headline accuracy, not hidden by it.
- Same numbers as in this talk: one source of truth.

Knowing when not to answer



- The model abstains when its predictive entropy is high.
- Threshold set on validation for 80% coverage; never tuned on test.
- Test: keeps 69% at **97.6%** accuracy; 2,206 images go to human review.
- Coverage drops at the new hospital: the model knows it is less sure.

Robustness check: corrupted inputs



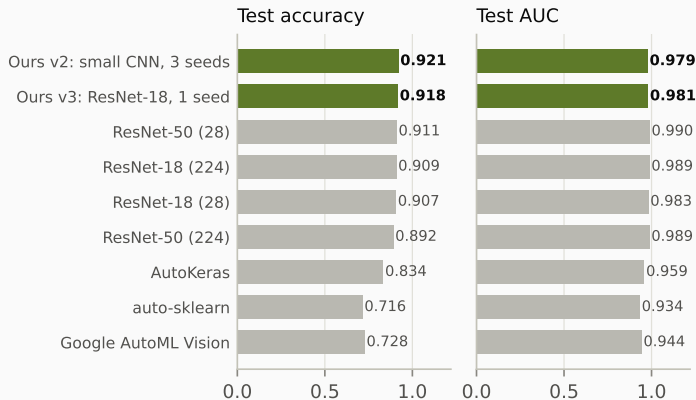
Test

- Gaussian pixel noise at three strengths.
- Question: does the abstention score separate noisy from clean images?

Result

- AUROC **0.943** on test (0.994 on validation).
- At most 0.2% of noisy images accepted.
- Limit: noise is a proxy; staining and scanner shifts are untested.

Comparison with the state of the art



Published: MedMNIST v2, Yang et al., *Sci. Data* 2023, Table 3 (mean of ≥ 3 runs).

Accuracy: **above every published baseline.**

AUC: slightly below the best ResNets.

Far ahead of AutoML tools, the closest kind of automated pipeline.

Acceptance verdict

Criterion	Status
Typed, checksummed hand-offs	✓
Zero direct agent calls	✓
Leakage audit before training	✓
Cited ideas	✓
Live web retrieval	✓
Validation-only selection	✓
Baseline comparison	✓
Representation ablations	✓
Seed variance (3 seeds)	✓

Criterion	Status
Abstention and human hand-off	✓
Reviewer report on every stage	✓
Decision provenance	✓
End-to-end run	✓
Fault detection $\geq 90\%$	●
Full-run replay	●
Approved risk mitigations	●
Independent TRL 7 sign-off	●

✓ evidenced in the project runs

● evidence file to be filed

13 / 17
criteria evidenced

4
need external evidence

Thank you

Maximum cohesion, minimum coupling.